Project Initialization and Planning Phase

Date	07 July 2026
Project Title	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

CaptionAI is an AI-powered web application designed to simplify social media content creation. It generates engaging captions, relevant hashtags, and effective call-to-action suggestions based on the user's input, helping users create platform-specific content quickly while maintaining creativity and consistency.

Project Overview	
Objective	The primary objective is to develop an AI-powered caption generation system that automatically produces engaging captions, relevant hashtags, and effective call-to-action suggestions for social media content, supporting multiple platforms through a simple and user-friendly interface.
Scope	The current version of CaptionAI focuses on generating captions, hashtags, and CTA suggestions for different social media platforms. It includes user authentication, caption history, and download functionality. Future versions may include multilingual support, image-based caption generation, content scheduling, and analytics.
Problem Statement	
Description	Creating engaging captions for social media platforms requires creativity, time, and knowledge of current trends. Many content creators, students, small businesses, and marketers struggle to write attractive captions and select relevant hashtags consistently. Existing tools are often expensive, limited in features, or require extensive manual editing.
Impact	Solving these issues will reduce the time required to create social media captions, improve the quality and creativity of content, help users maintain consistent social media engagement, and enhance productivity for individuals and businesses.
Proposed Solution	
Approach	CaptionAI provides an intelligent solution by using AI to generate high-quality captions, hashtags, and CTA suggestions within seconds. Users simply enter their topic, select the desired platform and tone, and the system produces ready-to-use content. It also saves caption history and allows users to copy or download the generated results.

Key Features	- AI-generated social media captions. - Platform-specific caption generation. - Tone selection for different writing styles. - Automatic hashtag generation. - Call-to-action (CTA) suggestions. - Caption history management. - Copy and download options. - Secure user authentication.

Objectives

- Develop an AI-powered caption generation system.
- Generate relevant hashtags automatically.
- Suggest effective call-to-action statements.
- Support multiple social media platforms.
- Save caption history for future reference.
- Provide a simple and user-friendly interface.

Target Users

- Content Creators
- Social Media Influencers
- Digital Marketers
- Students
- Small Business Owners
- Freelancers
- Startups

Expected Outcomes

- Reduce the time required to create social media captions.
- Improve the quality and creativity of content.
- Help users maintain consistent social media engagement.
- Provide a centralized platform for caption creation and management.
- Enhance productivity for individuals and businesses.

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU specifications, number of cores	Standard CPU (2 vCPUs) – no GPU required, as AI processing is handled by the Groq API
Memory	RAM specifications	2 GB RAM
Storage	Disk space for application, database, and logs	5 GB SSD

Resource Type	Description	Specification/Allocation
Software		
Frameworks	Python framework	Flask
Libraries	Additional libraries	Flask-SQLAlchemy, Flask-Login, Requests, python-dotenv
Development Environment	IDE	Visual Studio Code
Data		
Data	Source, size, format	No static training dataset is used. Content is generated dynamically via the Groq API based on user prompts; user accounts and caption history are stored in the SQLite database.